

Broom LED tube map — groups, Angios, ports, universes

Generated by [tube_map.py](#) — do not hand-edit. Change the knobs in that script and re-run.

136 tubes in **34 groups** of 4 across **5 Angio-8s**. One group = one data line = one Angio port = one sACN universe ($4 \times 40 = 160 \text{ px} = 480 \text{ ch} \leq 512$).

Chip **SM16703**, color order **RGB**, transport **sACN / E1.31**.

Angio placement (walking the open-front U)

Tubes numbered around the U from **front-left** → **back** → **front-right**: left

L01 (front)→ L56 (back), back B01 (left)→ B24 (right), right R01 (back)→ R56 (front).

Angio	Location	Tubes	Groups	Ports used	Spare ports
A1	Left-Front	28	G1–G7	7	1
A2	Left-Back	28	G8–G14	7	1
A3	Back	24	G15–G20	6	2
A4	Right-Back	28	G21–G27	7	1
A5	Right-Front	28	G28–G34	7	1

Full group map

Group	Angio	Port	Tubes	Px	Universe
G1	A1	1	L01 – L04	160	1
G2	A1	2	L05 – L08	160	2
G3	A1	3	L09 – L12	160	3
G4	A1	4	L13 – L16	160	4
G5	A1	5	L17 – L20	160	5
G6	A1	6	L21 – L24	160	6
G7	A1	7	L25 – L28	160	7
G8	A2	1	L29 – L32	160	8
G9	A2	2	L33 – L36	160	9
G10	A2	3	L37 – L40	160	10
G11	A2	4	L41 – L44	160	11
G12	A2	5	L45 – L48	160	12
G13	A2	6	L49 – L52	160	13
G14	A2	7	L53 – L56	160	14
G15	A3	1	B01 – B04	160	15
G16	A3	2	B05 – B08	160	16
G17	A3	3	B09 – B12	160	17
G18	A3	4	B13 – B16	160	18
G19	A3	5	B17 – B20	160	19
G20	A3	6	B21 – B24	160	20
G21	A4	1	R01 – R04	160	21

Group	Angio	Port	Tubes	Px	Universe
G22	A4	2	R05 – R08	160	22
G23	A4	3	R09 – R12	160	23
G24	A4	4	R13 – R16	160	24
G25	A4	5	R17 – R20	160	25
G26	A4	6	R21 – R24	160	26
G27	A4	7	R25 – R28	160	27
G28	A5	1	R29 – R32	160	28
G29	A5	2	R33 – R36	160	29
G30	A5	3	R37 – R40	160	30
G31	A5	4	R41 – R44	160	31
G32	A5	5	R45 – R48	160	32
G33	A5	6	R49 – R52	160	33
G34	A5	7	R53 – R56	160	34

Labeling scheme

Label every tube at **both ends** with its tube ID and its group. Example flag on a tube:

L05 / G2 = tube L05, group 2. Within a group the data chain runs **serpentine**: the tube nearest the Angio is the DIN head; flip alternate tubes so DOUT→DIN hops stay short, then reverse those strands in software.

Angio port –[330–470Ω]→ DIN[tube a] –DOUT→ DIN[tube b] → DIN[tube c]
 → DIN[tube d] (one group)

Install + test sequence

Do it **one group at a time**. Recommended order below finishes one Angio before moving to the next, so each controller + its network config is validated end-to-end. **Start with G15 (Back Angio A3)** as the very first group — it doubles as the open *bench-test-SM16703-on-Angio-8* check.

For each group: hang the 4 tubes → chain data (serpentine) → tap +24 V + shared GND → label both ends → ping me to light it from software → confirm all 4 tubes + color order → check the box.

A1 — LEFT-FRONT (28 TUBES)

- ☐ **G1** · port 1 · tubes L01–L04 · universe 1
- ☐ **G2** · port 2 · tubes L05–L08 · universe 2
- ☐ **G3** · port 3 · tubes L09–L12 · universe 3
- ☐ **G4** · port 4 · tubes L13–L16 · universe 4
- ☐ **G5** · port 5 · tubes L17–L20 · universe 5
- ☐ **G6** · port 6 · tubes L21–L24 · universe 6
- ☐ **G7** · port 7 · tubes L25–L28 · universe 7

A2 — LEFT-BACK (28 TUBES)

- ☐ **G8** · port 1 · tubes L29–L32 · universe 8
- ☐ **G9** · port 2 · tubes L33–L36 · universe 9
- ☐ **G10** · port 3 · tubes L37–L40 · universe 10
- ☐ **G11** · port 4 · tubes L41–L44 · universe 11
- ☐ **G12** · port 5 · tubes L45–L48 · universe 12
- ☐ **G13** · port 6 · tubes L49–L52 · universe 13
- ☐ **G14** · port 7 · tubes L53–L56 · universe 14

A3 — BACK (24 TUBES)

- ☐ **G15** · port 1 · tubes B01–B04 · universe 15
- ☐ **G16** · port 2 · tubes B05–B08 · universe 16
- ☐ **G17** · port 3 · tubes B09–B12 · universe 17

- ☐ **G18** · port 4 · tubes B13–B16 · universe 18
- ☐ **G19** · port 5 · tubes B17–B20 · universe 19
- ☐ **G20** · port 6 · tubes B21–B24 · universe 20

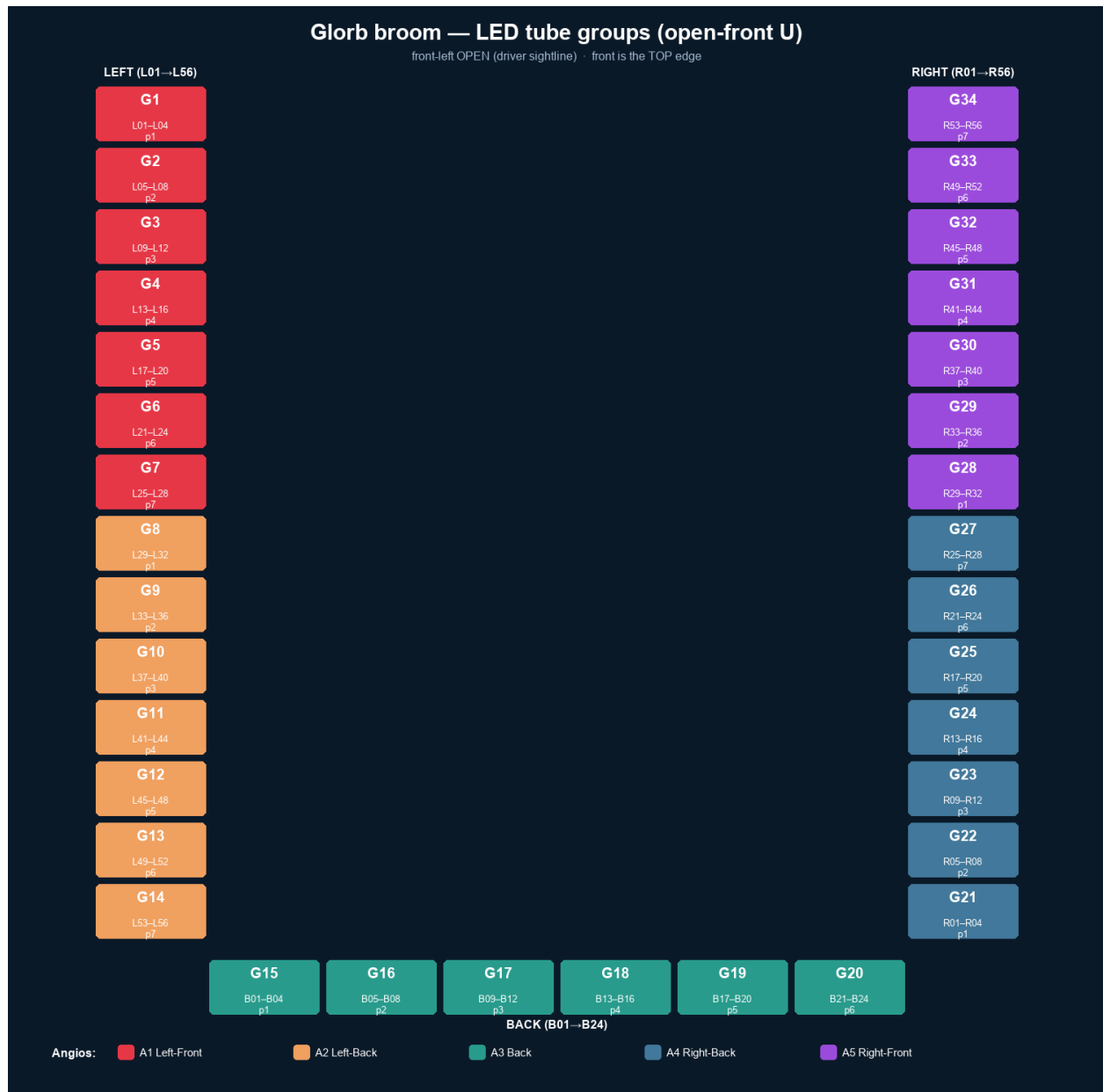
A4 — RIGHT-BACK (28 TUBES)

- ☐ **G21** · port 1 · tubes R01–R04 · universe 21
- ☐ **G22** · port 2 · tubes R05–R08 · universe 22
- ☐ **G23** · port 3 · tubes R09–R12 · universe 23
- ☐ **G24** · port 4 · tubes R13–R16 · universe 24
- ☐ **G25** · port 5 · tubes R17–R20 · universe 25
- ☐ **G26** · port 6 · tubes R21–R24 · universe 26
- ☐ **G27** · port 7 · tubes R25–R28 · universe 27

A5 — RIGHT-FRONT (28 TUBES)

- ☐ **G28** · port 1 · tubes R29–R32 · universe 28
- ☐ **G29** · port 2 · tubes R33–R36 · universe 29
- ☐ **G30** · port 3 · tubes R37–R40 · universe 30
- ☐ **G31** · port 4 · tubes R41–R44 · universe 31
- ☐ **G32** · port 5 · tubes R45–R48 · universe 32
- ☐ **G33** · port 6 · tubes R49–R52 · universe 33
- ☐ **G34** · port 7 · tubes R53–R56 · universe 34

Diagram



Related

- [controllers.md](#) — data wiring, shared-ground, serpentine reverse
- [led-tubes.md](#) — SM16703 electricals (16 px/m, 5 V data, RGB, series resistor)
- [../electrical/led-wiring.md](#) — power injection (separate from data)
- [tube-map.json](#) — machine-readable map for the control software